# Active Matter × MPM: Cells in a Deformable Blob

A Plexus prototype toward embryogenesis / morphogenesis

Plexus prototype `prototype/embryogenesis`

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**Abstract**

We couple *active-matter agents* (self-propelled, aligning "cells") to a *Material Point Method* (MLS-MPM) continuum "blob" of water or elastic tissue. The two subsystems talk through the shared MPM background grid: the material *drags* the cells and its surface-tension interface *confines* them; the cells scatter momentum back and *deform* the material. The blob rotates slowly. Everything is built from Plexus operators; the only new physics is three small operators (`mpm_to_agent`, `agent_to_mpm`, `mpm_spin`). We show stable 2D prototypes for a water disc, an elastic disc, and a four-cell-type showcase, and document the confinement ("escape") failure and its fix. The system is the substrate for an agentic exploration loop that maps its morphogenetic regimes.


## 1 Goal

Multi-type cells migrate collectively inside a large disc of MPM material (water or elastic). The material drags the cells; the cells create stress/deformation in the material; the cells cannot leave the disc, which is held together by surface tension. Ultimately we want to reproduce embryogenesis / morphogenesis dynamics. This note records the working prototypes and the code they required.

## 2 Model

Two node sets share one world and one clock.

**Cells (active matter).** A set `agent` of self-propelled particles carrying a unit heading $\mathbf{n}_i$. Motion is heading-kinematic (overdamped, first order): the heading is steered by polar alignment (Vicsek), chemotaxis, and angular noise, while `glide` moves each cell along its heading at speed $v_0^{(\tau_i)}$ (per type $\tau_i$) and `repel` enforces a hard core:

$$\dot{\mathbf{r}}_i = v_0^{(\tau_i)}\,\mathbf{n}_i + \sum_j \mathbf{f}_{ij}, \qquad \dot{\mathbf{n}}_i = \Gamma \sum_{j\in\partial i} \frac{\mathbf{n}_j - (\mathbf{n}_j\cdot\mathbf{n}_i)\mathbf{n}_i}{r_{ij}} + \omega\,\widehat{\nabla c} + \sqrt{2D_r}\,\xi_i. \tag{1}$$

**Material (MLS-MPM).** A disc of material points (set `mpm_particle`, child of a `cell`) with deformation gradient $F$ and affine velocity $C$. The decomposed solver is the standard cycle `mpm_strain` → `p2g` → `mpm_grid_update` → `g2p`, run for $\lfloor \Delta t/\Delta t_{\mathrm{sub}} \rfloor$ substeps per frame. Water is a liquid band ($\mu = 0$, pressure only); elastic tissue is fixed-corotated with a thin liquid *skin*. The liquid colour field $c$ drives the continuum surface force (CSF) that holds the blob's boundary.

**Two-way coupling (new).** Both directions route through the MPM grid velocity $\mathbf{v}_g$ and mass/momentum $m_g$, $\mathbf{p}_g$:

$$\text{material} \to \text{cell:} \quad \dot{\mathbf{r}}_i \mathrel{+}= k\,\mathbf{v}_g(\mathbf{r}_i) \;+\; \kappa\,\nabla c(\mathbf{r}_i) \qquad (\texttt{mpm\_to\_agent}) \tag{2}$$

$$\text{cell} \to \text{material:} \quad m_g \mathrel{+}= \textstyle\sum_i w\,\mu_a, \quad \mathbf{p}_g \mathrel{+}= \textstyle\sum_i w\,\mu_a v_0 \mathbf{n}_i \qquad (\texttt{agent\_to\_mpm}) \tag{3}$$

using the *same* quadratic B-spline weights $w$ MPM uses for its own transfers. In (2) the first term is fluid drag (the cell is advected by the flow, $k$=1 = fully carried) and the second is the **surface-tension confinement**: $c$ is $\sim 1$ inside the fluid and $\sim 0$ outside, so $\nabla c$ points *inward* at the interface and pushes cells back in — a soft membrane, not a hard wall. In (3) cells have no physical mass, so an effective mass $\mu_a$ (times a gain $k$) sets the push; scattering onto $m_g, \mathbf{p}_g$ *after* `p2g` but *before* `mpm_grid_update` makes the grid solve on the combined (material + cell) momentum, so `g2p` carries the cells' push into real deformation. Rotation is `mpm_spin`: a body force driving the disc toward solid-body rotation $\mathbf{v}_{\text{rot}} = \omega\,\hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{c})$.

**One clock.** The engine integrates on a single global $\Delta t$; we make the MPM clock master ($\Delta t = 2\times10^{-3}$, $\Delta t_{\text{sub}} = 2\times10^{-4}$) and express the cell gains in that time unit.

## 3 Results

All panels: cells are coloured dots by type, the MPM material is the translucent blue blob, black background. Time runs left→right.

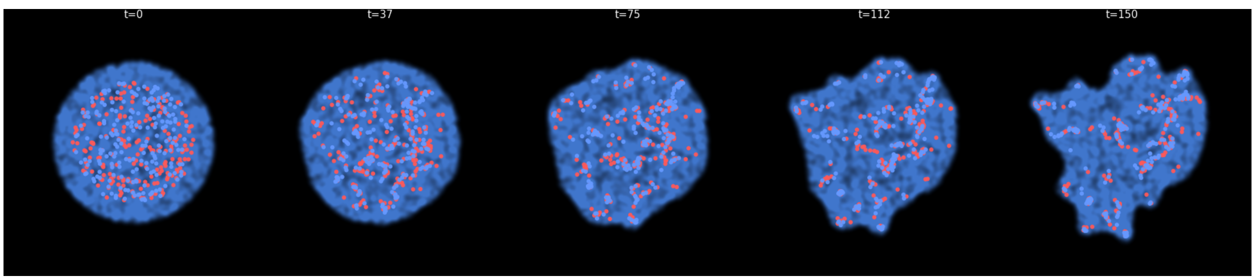


Figure 1: **Water disc.** Two cell types swim in a cohesive rotating water blob, form aligned streams, and gently lobe the boundary. 0% escape.

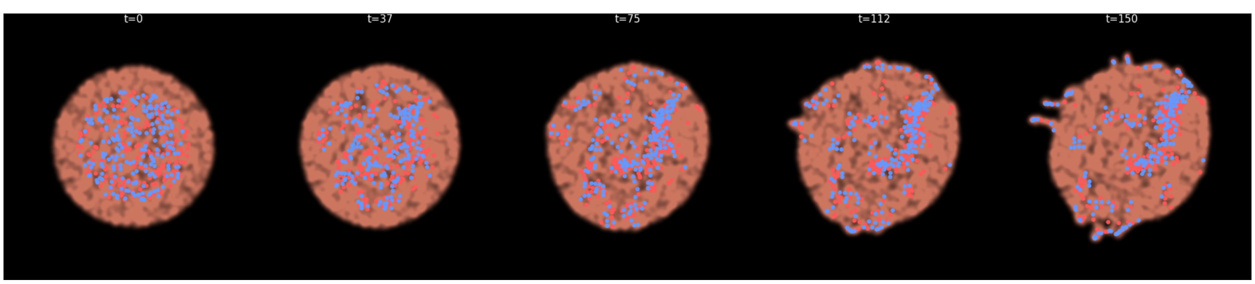


Figure 2: **Elastic disc.** A deformable tissue with a liquid skin holds a rounder shape; a coherent stream of cells (polar order ≈ 0.46) migrates across it. 0.25% escape.

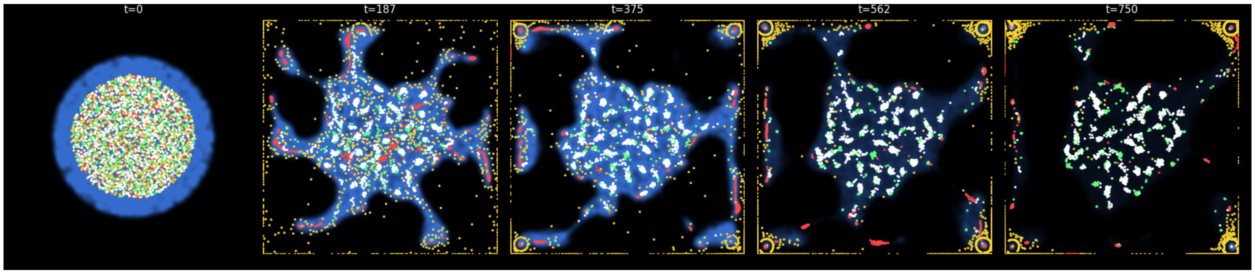


Figure 3: **Four-type showcase** (4000 cells). Per-type behaviour via masked operator instances: a (red) flocker, b (gold) disperser, c (green) chemotactic aggregator, d (white) noisy. All confined and carried by the rotating water blob.

## 4 The escape problem and its fix

The first attempt let nearly all cells leak out and the water disc foam apart (Fig. 4). Two coupled causes: (i) an under-resolved liquid (too few particles per grid cell) with *zero* viscous drag, agitated by spin and cell push, fragments; (ii) once the interface is ragged and the confinement gain is weak, cells walk straight through it. The fix is a combination, summarised in Table 1: fewer grid cells (more particles/cell), viscous drag, stronger surface tension, gentler spin/push, and a much stronger confinement gain.

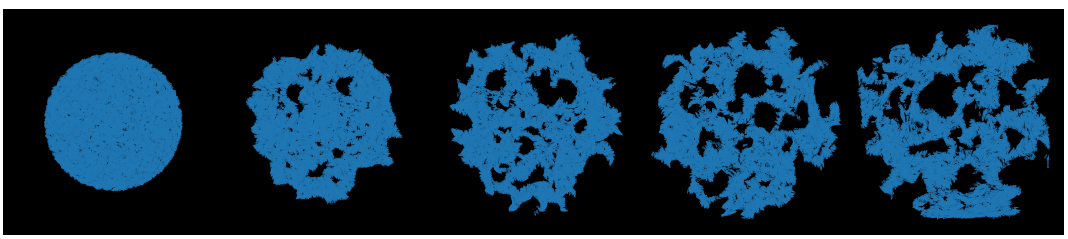

Figure 4: **Failure kept on purpose (v1).** The water disc foams, fragments and spreads; cells escape to the walls.

| run | escaped | disc growth | aniso | polar | key change |
|---|---|---|---|---|---|
| water v1 (fail) | $\sim 0.95$ | 0.30 | 0.60 | — | baseline: n_grid 96, drag 0, confine 30 |
| water v2 | 0.025 | 0.042 | 0.14 | 0.14 | n_grid 64, drag 0.25, ST 120, confine 70 |
| water v3 | **0.000** | 0.057 | 0.14 | 0.14 | drag 0.4, ST 160–300, confine 140 |
| elastic v3 | 0.003 | 0.049 | 0.08 | **0.46** | elastic core + liquid skin |

Table 1: Confinement / stability sweep. "escaped" = fraction of cells outside the blob at the end; "aniso" = blob boundary roughness (0 = perfect disc).

## 5 Compute budget

On one RTX A6000, the four-type showcase (4000 cells, 16,000 material points, $n_{\text{grid}} = 64$) simulates 1500 frames in $\approx 100\,\text{s}$. This leaves ample headroom for the $\leq 20$ min-per-job L4 budget of the agentic exploration loop.

## 6 Next: agentic exploration

The prototype is the substrate for an autonomous loop (a clone of `prototype/active_matter2`) in which an agent proposes parameter variations, runs 8 jobs in parallel on L4, and maps the morphogenetic regimes (mixing vs. segregation, blob rounding vs. lobing, engulfment, rotation-shear banding), with a batch montage as the visual feedback. In 3D the same operators drive a $2\times2$ summary movie: cell trajectories with tails, a transparent deformation volume, the affine-velocity field, and a combined view.